

Tech & Teach

Digital Chip Design Program

RTL Design

Task #06

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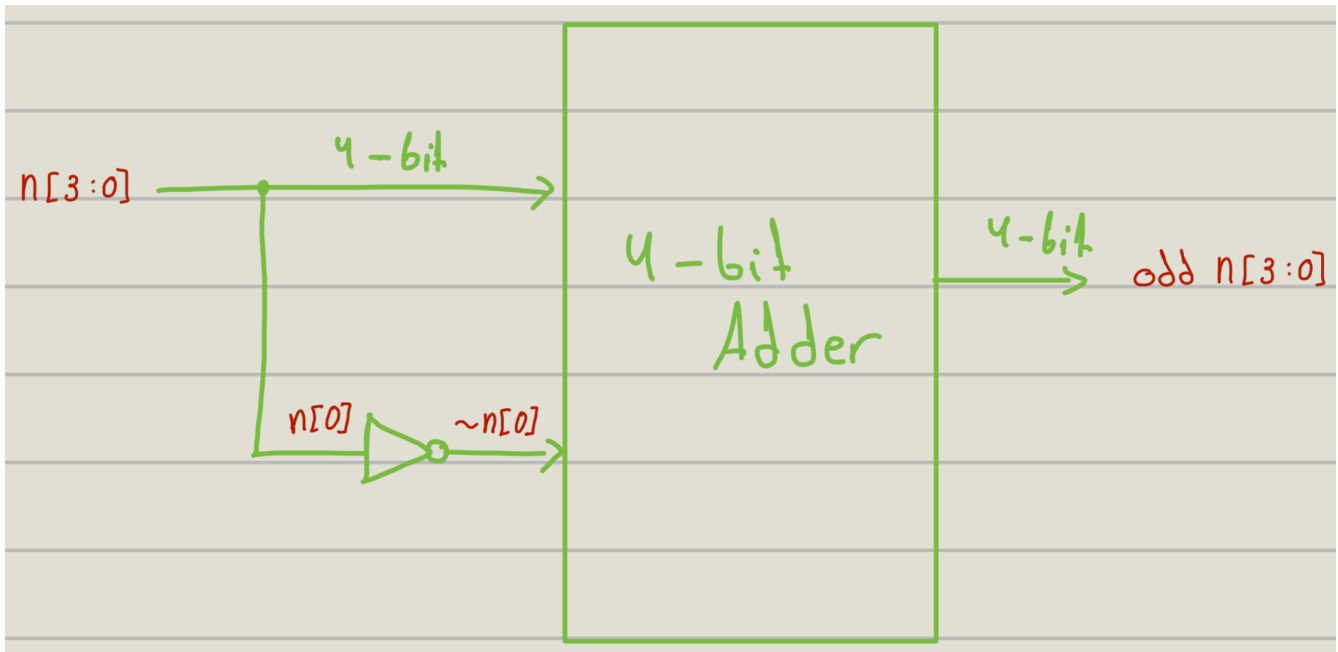
Combinational Datapath Design Part 2

Version 1.0

Task #1.1

1.1.1.2

Microarchitecture

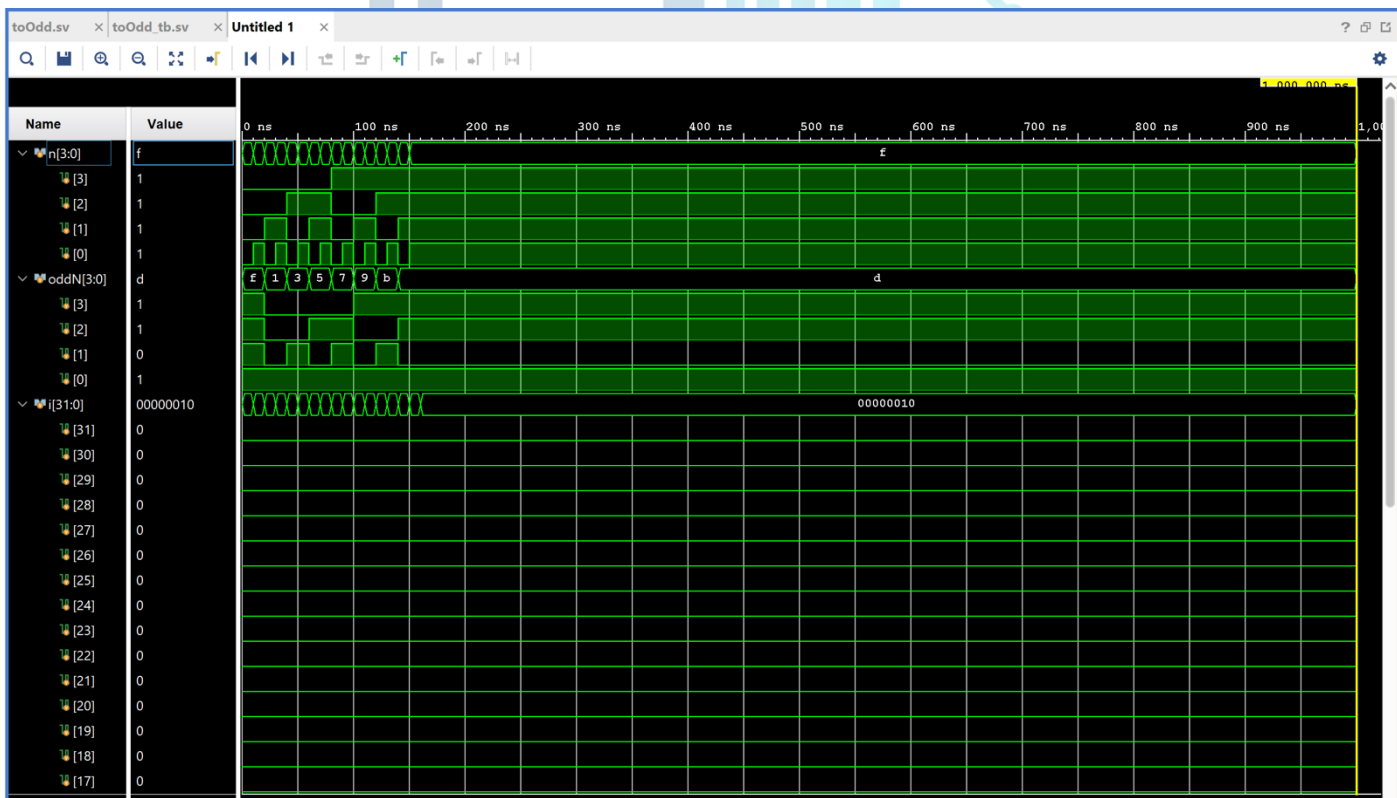


1.1.1.3

```
1 module toOdd (  
2     input logic [3:0] n,  
3     output logic [3:0] oddN  
4 );  
5  
6     assign oddN = n + (~n[0]);  
7  
8 endmodule
```

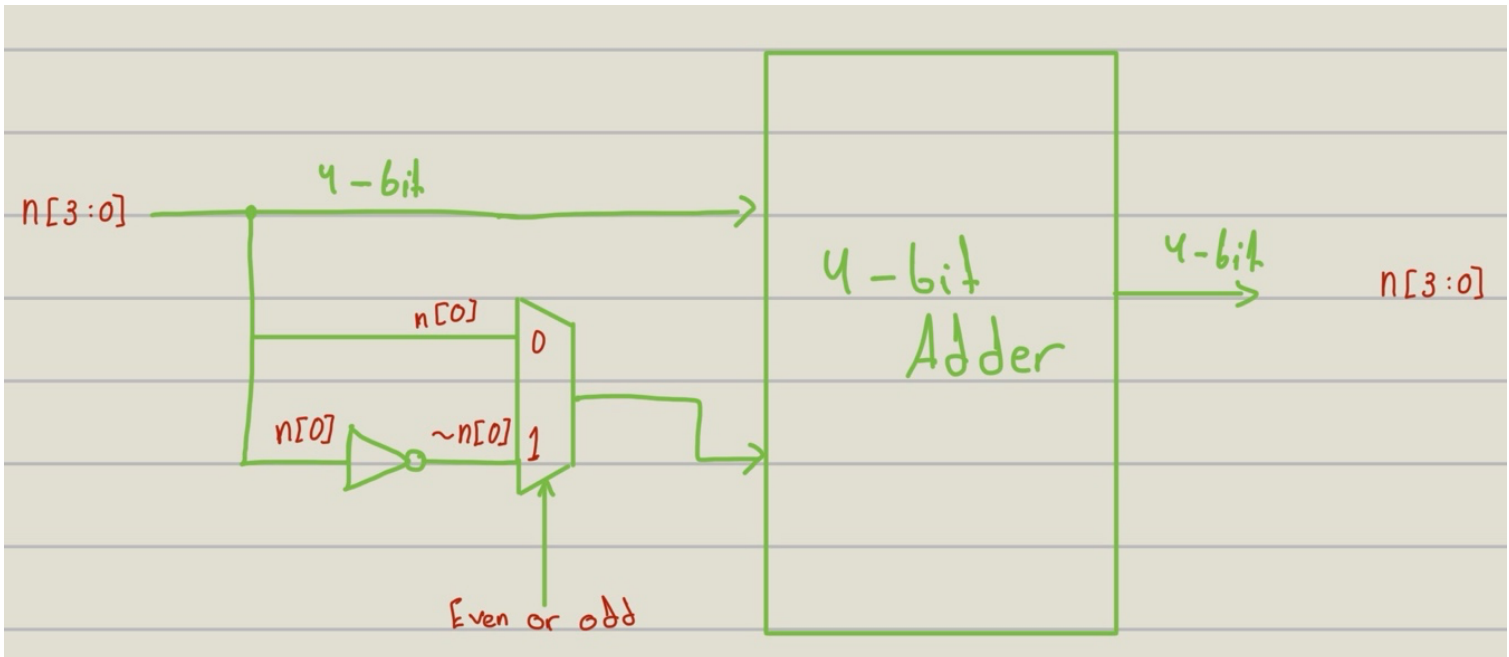
1.1.1.4

```
1  `timescale 1ns / 1ps
2  module toOdd_tb();
3      logic [3:0] n;
4      logic [3:0] oddN;
5
6      toOdd dut (
7          .n(n),
8          .oddN(oddN)
9      );
10
11     int i;
12     initial begin
13         for (i = 0; i < 16; i = i + 1) begin
14             n = i;
15             #10;
16         end
17         #10 $finish;
18     end
19 endmodule
```



1.1.2.2

Microarchitecture

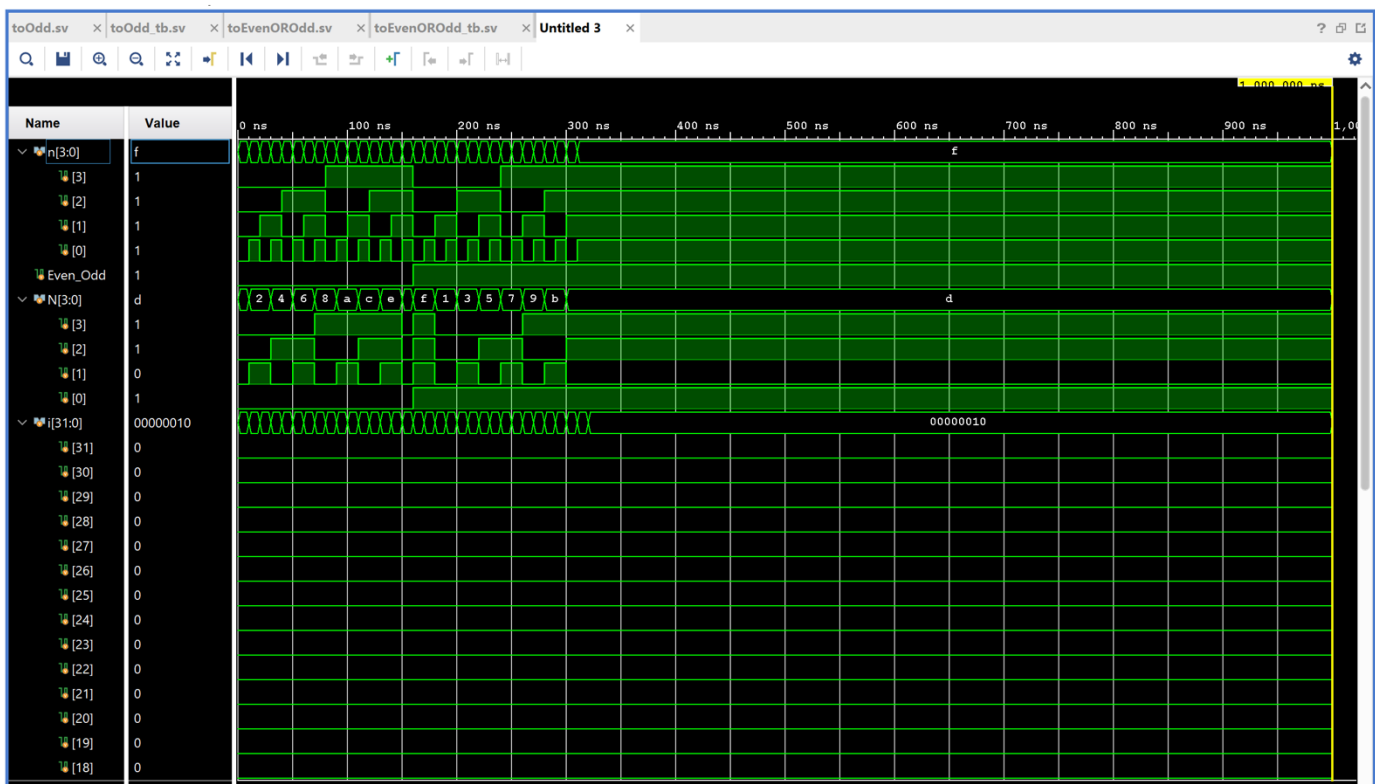


1.1.2.3

```
1 module toEvenOROdd (  
2     input logic [3:0] n,  
3     input logic      Even_Odd,  
4     output logic [3:0] N  
5 );  
6  
7     assign N = n + (Even_Odd ? (~n[0]) : n[0]);  
8  
9 endmodule
```

1.1.2.4

```
1  `timescale 1ns / 1ps
2
3  module toEvenOROdd_tb;
4      logic [3:0] n;
5      logic      Even_Odd;
6      logic [3:0] N;
7
8      toEvenOROdd dut (
9          .n(n),
10         .Even_Odd(Even_Odd),
11         .N(N)
12     );
13
14     int i;
15     initial begin
16         Even_Odd = 0;
17         for (i = 0; i < 16; i = i + 1) begin
18             n = i;
19             #10;
20         end
21
22         Even_Odd = 1;
23         for (i = 0; i < 16; i = i + 1) begin
24             n = i;
25             #10;
26         end
27
28         #10 $finish;
29     end
30 endmodule
```



Task #1.2

1.2.1.1

```

2      input logic [N-1:0] number,
3      output logic  special);

```

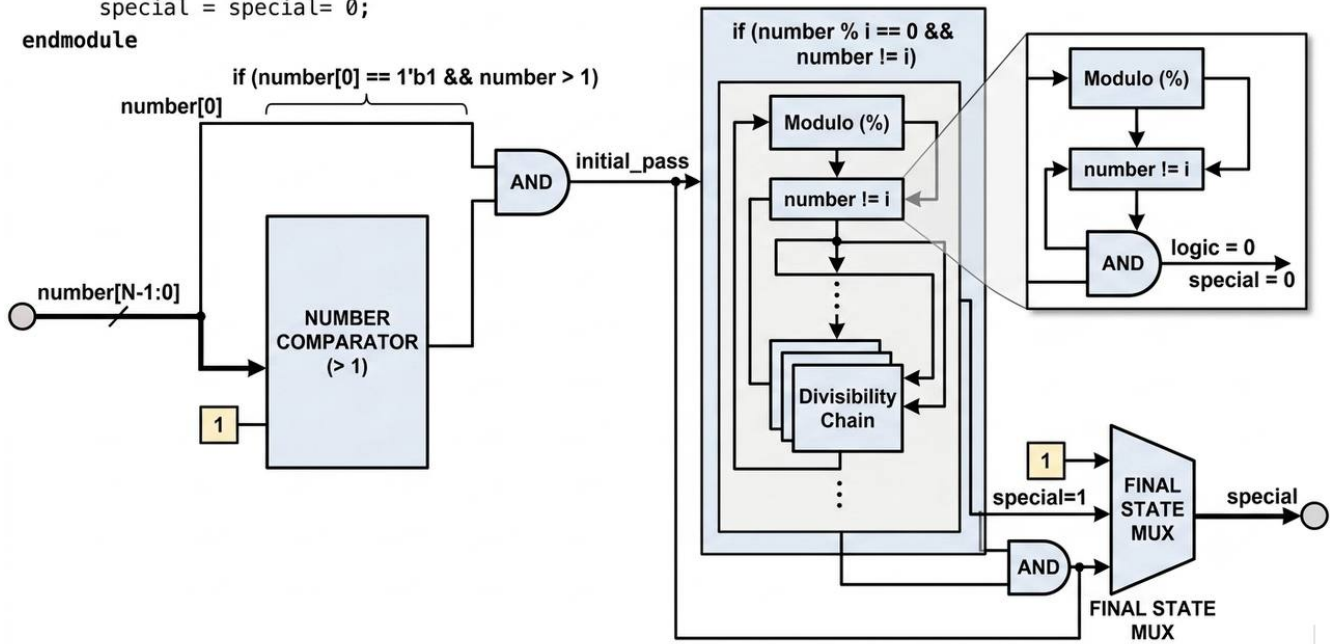
1.2.1.2

Microarchitecture

```

module isSpecial(number[N-1:0], starg, N-1:0));
  if (number[0] == 1'b1 && number > 1)
    special = special= 0;
endmodule

```

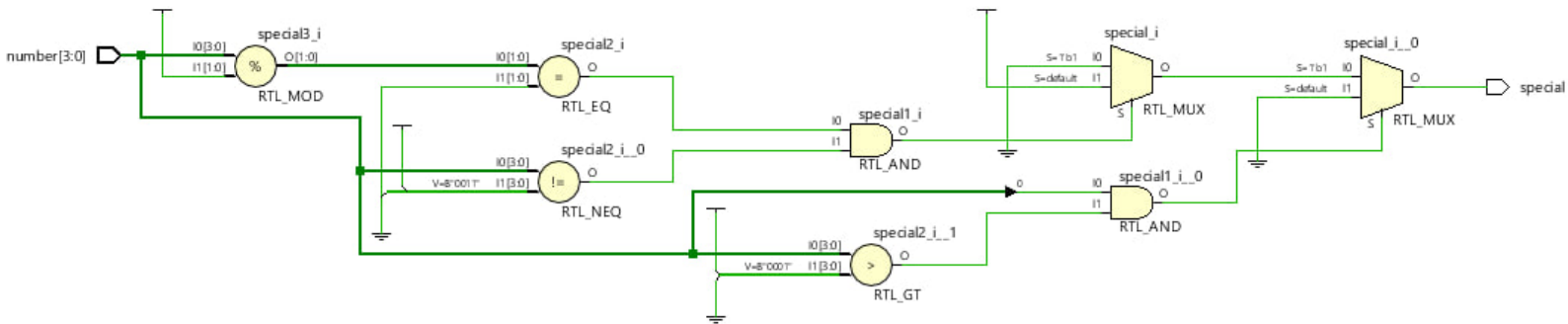


1.2.1.3

```

1 module isSpecial #(parameter N = 4) (
2     input logic [N-1:0] number,
3     output logic  special);
4
5     always_comb begin
6         special = 1'b0;
7
8         if (number[0] == 1'b1 && number > 1) begin
9             special = 1'b1;
10            for (int i = 3; i * i <= (1 << N); i += 2) begin
11                if (number % i == 0 && number != i) begin
12                    special = 1'b0;
13                end
14            end
15        end
16    end
17
18 endmodule

```



1.2.1.4

Project Summary x isspecial_tb.sv x Schematic x

C:/Users/farah/Task2Day7/Task2Day7.srscs/sim_1/new/isspecial_tb.sv

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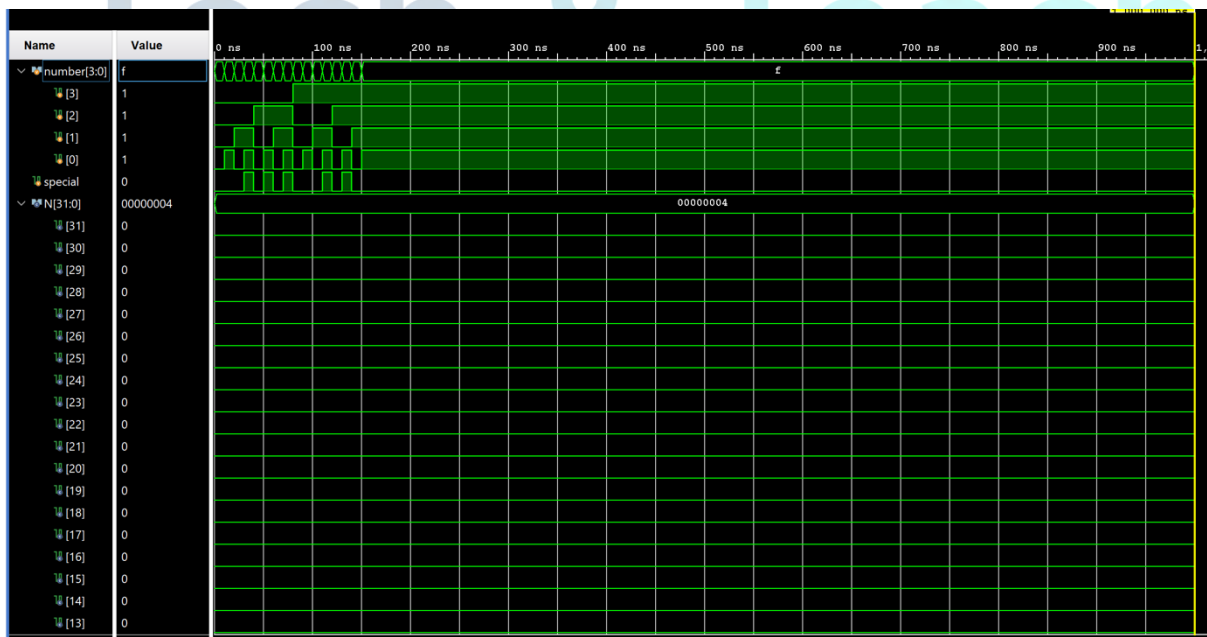
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```

1  `timescale 1ns / 1ps
2  module isspecial_tb;
3      parameter N = 4;
4      logic [N-1:0] number;
5      logic          special;
6
7      isSpecial #(N) uut (
8          .number(number),
9          .special(special)
10     );
11
12     initial begin
13         for (int i = 0; i < (1 << N); i++) begin
14             number = i;
15             #10;
16         end
17         $finish;
18     end
19 endmodule

```

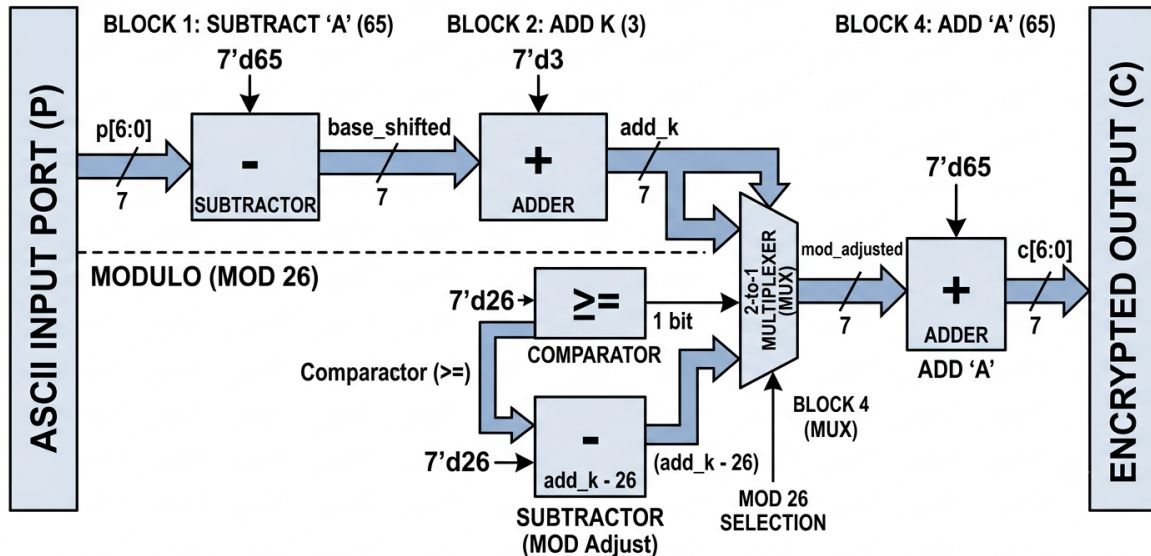


Task #1.3

1.3.1.2

Microarchitecture

CAESAR CIPHER MICROARCHITECTURE (k=3 for ASCII A-Z)



1.3.1.3

```
1 module caesar_cipher (  
2     input logic [6:0] p,  
3     output logic [6:0] c  
4 );  
5  
6     logic [6:0] base_shifted;  
7     logic [6:0] add_k;  
8     logic [6:0] mod_adjusted;  
9  
10    always_comb begin  
11        base_shifted = p - 7'd65;  
12        add_k = base_shifted + 7'd3;  
13  
14        if (add_k >= 7'd26) begin  
15            mod_adjusted = add_k - 7'd26;  
16        end else begin  
17            mod_adjusted = add_k;  
18        end  
19  
20        c = mod_adjusted + 7'd65;  
21    end  
22  
23 endmodule
```


1.3.1.4

```
1  `timescale 1ns / 1ps
2  module caesar_cipher_tb();
3
4      logic [6:0] p;
5      logic [6:0] c;
6
7      caesar_cipher dut (
8          .p(p),
9          .c(c)
10     );
11
12     int i;
13
14     initial begin
15         for (i = 65; i <= 90; i = i + 1) begin
16             p = i;
17             #10;
18         end
19
20         #50;
21         #10 $finish;
22     end
23
24 endmodule
```



Work distribution

Section	Farah&Joud	Tala&Horiah	Rana&Hoor&Sandra
Report	✓	✓	✓
Task 1.1			✓
Task 1.2	✓		
Task 1.3		✓	